symcoder parity-blindness demo

shadow event #0, parity = -1

1 About this document

This document is one half of a parity-blindness demonstration. A companion document, identical in structure but built from the parity-flipped event, exists alongside it. Their Phase 1 and Phase 2 encoded values should match exactly, despite the events lying in distinct $SO(3)$ orbits.

This run uses event with parity sign: **This run uses event with parity sign:** -1

Shadow event index: **Shadow event index:** 0 (from `symcoder.tests._parity_shadow_events.SHADOW_EVENT`

Companion document: **Companion document:** `demo_shadow_event_parity_+1.tex`

2 Setup

Particle types: electrons $= \{a, b, c\}$, muons $= \{p, q\}$
Group G: $G = S_{\text{electrons}} \times S_{\text{muons}} = S_3 \times S_2$, order $= 12$

Operations:
$|\mathbf{x}|$ (rank 1, symmetric)
$\mathbf{x} \cdot \mathbf{y}$ (rank 2, symmetric)
$\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (rank 3, antisymmetric)

Event vectors (3-D), parity sign $= -1$:
a: $\mathbf{a} = (1.00,\ 1.00,\ 1.00)$

b: $\mathbf{b} = (1.00,\ 1.00,\ -0.00)$

c: $\mathbf{c} = (-1.00,\ 1.00,\ 1.00)$

p: $\mathbf{p} = (-0.00,\ -1.00,\ -0.00)$

q: $\mathbf{q} = (-0.00,\ 1.00,\ -0.00)$

3 Phase 1 Encoding (one orbit per FlavouredOperator in repS)

8 FlavouredOperators in repS

3.1 FlavouredOperator: $|\mathbf{x}|$ flavour $= (1, 0)$

Canonical representative: $|\mathbf{a}|$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $|\mathbf{a}|$, $|\mathbf{b}|$, $|\mathbf{c}|$
Eval values: +1.7321, +1.4142, +1.7321
Encoded [3 reals]: +1.4142, +1.7321, +1.7321

3.2 FlavouredOperator: $|\mathbf{x}|$ flavour $= (0, 1)$

Canonical representative: $|\mathbf{p}|$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $|\mathbf{p}|$, $|\mathbf{q}|$
Eval values: +1.0000, +1.0000
Encoded [2 reals]: +1.0000, +1.0000

3.3 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (2, 0)$

Canonical representative: $\mathbf{a} \cdot \mathbf{b}$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$, $\mathbf{b} \cdot \mathbf{c}$
Eval values: +2.0000, +1.0000, +0.0000
Encoded [3 reals]: +0.0000, +1.0000, +2.0000

3.4 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (1, 1)$

Canonical representative: $\mathbf{a} \cdot \mathbf{p}$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{q}$
Eval values: -1.0000, +1.0000, -1.0000, +1.0000, -1.0000, +1.0000
Encoded [6 reals]: -1.0000, -1.0000, -1.0000, +1.0000, +1.0000, +1.0000

3.5 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (0, 2)$

Canonical representative: $\mathbf{p} \cdot \mathbf{q}$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{p} \cdot \mathbf{q}$
Eval values: -1.0000
Encoded [1 reals]: -1.0000

3.6 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (3, 0)$

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$
Encoder: `HalfSortEncoder` — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$
Eval values: +2.0000
Encoded [1 reals]: +2.0000

3.7 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (2, 1)$

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$
Encoder: `HalfSortEncoder` — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$
Eval values: -1.0000, +1.0000, +2.0000, -2.0000, +1.0000, -1.0000
Encoded [6 reals]: +1.0000, +1.0000, +1.0000, +1.0000, +2.0000, +2.0000

3.8 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (1, 2)$

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$
Encoder: `HalfSortEncoder` — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$
Eval values: +0.0000, +0.0000, +0.0000
Encoded [3 reals]: +0.0000, +0.0000, +0.0000

4 Phase 2 Encoding (OverlapBlocks, complementarity-drop disabled)

4.1 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 2
Encoded values: +2.8284, +2.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ |\mathbf{q}|) \to (+1.0000,\ +1.0000)$
$(|\mathbf{q}|,\ |\mathbf{p}|) \to (+1.0000,\ +1.0000)$
PairFlavour overlap=(0, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(0, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(|\mathbf{p}|,\ |\mathbf{p}|) \to (+1.0000,\ +1.0000)$
$(|\mathbf{q}|,\ |\mathbf{q}|) \to (+1.0000,\ +1.0000)$

4.2 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +9.7566, -24.5959, +48.7832, -138.3837, +12.0136, -118.7878, -159.0918, +55.1918, -140.8499, +47.1918, -8.7710
Decoded pairs (u, v):
$(|\mathbf{p}|,\ |\mathbf{a}|) \to (+1.0000,\ +1.7321)$
$(|\mathbf{q}|,\ |\mathbf{a}|) \to (+1.0000,\ +1.7321)$
$(|\mathbf{p}|,\ |\mathbf{b}|) \to (+1.0000,\ +1.7321)$
$(|\mathbf{q}|,\ |\mathbf{b}|) \to (+1.0000,\ +1.7321)$
$(|\mathbf{p}|,\ |\mathbf{c}|) \to (+1.0000,\ +1.4142)$
$(|\mathbf{q}|,\ |\mathbf{c}|) \to (+1.0000,\ +1.4142)$

4.3 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.0000, -2.0000, +0.0000, -2.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{p} \cdot \mathbf{q}) \rightarrow (+1.0000,\ -1.0000)$
$(|\mathbf{q}|,\ \mathbf{p} \cdot \mathbf{q}) \rightarrow (+1.0000,\ -1.0000)$

4.4 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +0.0000, +18.0000, +0.0000, +32.0000, +0.0000, +36.0000, +0.0000, +24.0000, +0.0000, +8.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a} \cdot \mathbf{q}) \rightarrow (+1.0000,\ -1.0000)$
$(|\mathbf{q}|,\ \mathbf{a} \cdot \mathbf{p}) \rightarrow (+1.0000,\ -1.0000)$
$(|\mathbf{p}|,\ \mathbf{b} \cdot \mathbf{q}) \rightarrow (+1.0000,\ -1.0000)$
$(|\mathbf{q}|,\ \mathbf{b} \cdot \mathbf{p}) \rightarrow (+1.0000,\ +1.0000)$
$(|\mathbf{p}|,\ \mathbf{c} \cdot \mathbf{q}) \rightarrow (+1.0000,\ +1.0000)$
$(|\mathbf{q}|,\ \mathbf{c} \cdot \mathbf{p}) \rightarrow (+1.0000,\ +1.0000)$
PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +0.0000, +18.0000, +0.0000, +32.0000, +0.0000, +36.0000, +0.0000, +24.0000, +0.0000, +8.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a} \cdot \mathbf{p}) \rightarrow (+1.0000,\ -1.0000)$
$(|\mathbf{q}|,\ \mathbf{a} \cdot \mathbf{q}) \rightarrow (+1.0000,\ -1.0000)$
$(|\mathbf{p}|,\ \mathbf{b} \cdot \mathbf{p}) \rightarrow (+1.0000,\ -1.0000)$
$(|\mathbf{q}|,\ \mathbf{b} \cdot \mathbf{q}) \rightarrow (+1.0000,\ +1.0000)$
$(|\mathbf{p}|,\ \mathbf{c} \cdot \mathbf{p}) \rightarrow (+1.0000,\ +1.0000)$
$(|\mathbf{q}|,\ \mathbf{c} \cdot \mathbf{q}) \rightarrow (+1.0000,\ +1.0000)$

4.5 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +6.0000, +2.0000, +30.0000, -32.0000, +48.0000, -59.0000, +24.0000, -38.0000, -6.0000, -8.0000, -6.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a} \cdot \mathbf{b}) \rightarrow (+1.0000,\ +2.0000)$
$(|\mathbf{q}|,\ \mathbf{a} \cdot \mathbf{b}) \rightarrow (+1.0000,\ +2.0000)$
$(|\mathbf{p}|,\ \mathbf{a} \cdot \mathbf{c}) \rightarrow (+1.0000,\ +1.0000)$
$(|\mathbf{q}|,\ \mathbf{a} \cdot \mathbf{c}) \rightarrow (+1.0000,\ +1.0000)$
$(|\mathbf{p}|,\ \mathbf{b} \cdot \mathbf{c}) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ \mathbf{b} \cdot \mathbf{c}) \rightarrow (+1.0000,\ +0.0000)$

4.6 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12

Encoded values: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to$ (+1.0000, +0.0000)
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to$ (+1.0000, +0.0000)
$(|\mathbf{p}|,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to$ (+1.0000, +0.0000)
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to$ (+1.0000, −0.0000)
$(|\mathbf{p}|,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to$ (+1.0000, −0.0000)
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to$ (+1.0000, −0.0000)

4.7 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +12.0000, +78.0000, +340.0000, +1089.0000, +2664.0000, +5072.0000, +7536.0000, +8664.0000, +7520.0000, +4704.0000, +1920.0000, +400.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to$ (+1.0000, +2.0000)
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to$ (+1.0000, +2.0000)
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to$ (+1.0000, −2.0000)
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to$ (+1.0000, −2.0000)
$(|\mathbf{p}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to$ (+1.0000, −1.0000)
$(|\mathbf{q}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to$ (+1.0000, −1.0000)
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to$ (+1.0000, −1.0000)
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to$ (+1.0000, −1.0000)
... (4 more pairs) PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +12.0000, +78.0000, +340.0000, +1089.0000, +2664.0000, +5072.0000, +7536.0000, +8664.0000, +7520.0000, +4704.0000, +1920.0000, +400.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to$ (+1.0000, +2.0000)
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to$ (+1.0000, +2.0000)
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to$ (+1.0000, −2.0000)
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to$ (+1.0000, −2.0000)
$(|\mathbf{p}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to$ (+1.0000, −1.0000)
$(|\mathbf{q}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to$ (+1.0000, −1.0000)
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to$ (+1.0000, −1.0000)
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to$ (+1.0000, −1.0000)
... (4 more pairs)

4.8 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=4 **PairFlavour** overlap= $(0,0)$, type=`12`, kind=`ASSOC`, encoded dim= 4
Encoded values: +4.0000, +14.0000, +20.0000, +25.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to$ (+1.0000, +2.0000)
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to$ (+1.0000, +2.0000)
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to$ (+1.0000, −2.0000)
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to$ (+1.0000, −2.0000)

4.9 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
Encoded values: +13.7980, +79.3939, +243.8684, +421.7673, +389.4439, +150.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ |\mathbf{b}|) \to (+1.4142,\ +1.7321)$
$(|\mathbf{a}|,\ |\mathbf{c}|) \to (+1.4142,\ +1.7321)$
$(|\mathbf{b}|,\ |\mathbf{a}|) \to (+1.7321,\ +1.7321)$
$(|\mathbf{b}|,\ |\mathbf{c}|) \to (+1.7321,\ +1.7321)$
$(|\mathbf{c}|,\ |\mathbf{a}|) \to (+1.7321,\ +1.4142)$
$(|\mathbf{c}|,\ |\mathbf{b}|) \to (+1.7321,\ +1.4142)$
PairFlavour overlap=(1, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(|\mathbf{a}|,\ |\mathbf{a}|) \to (+1.4142,\ +1.4142)$
$(|\mathbf{b}|,\ |\mathbf{b}|) \to (+1.7321,\ +1.7321)$
$(|\mathbf{c}|,\ |\mathbf{c}|) \to (+1.7321,\ +1.7321)$

4.10 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: +4.8783, -3.0000, +4.8990, -9.7566, -0.6357, -6.8990
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{p} \cdot \mathbf{q}) \to (+1.7321,\ -1.0000)$
$(|\mathbf{b}|,\ \mathbf{p} \cdot \mathbf{q}) \to (+1.7321,\ -1.0000)$
$(|\mathbf{c}|,\ \mathbf{p} \cdot \mathbf{q}) \to (+1.4142,\ -1.0000)$

4.11 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +19.5133, +0.0000, +180.3837, +0.0000, +1041.3172, +0.0000, +4173.3591, +0.0000, +12220.0778, +0.0000, +26792.4569, -0.0000, +44315.6341, -0.0000, +54903.0525, -0.0000, +49729.9824, -0.0000, +31305.6650, -0.0000, +12325.7542, +0.0000, +2304.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{b} \cdot \mathbf{p}) \to (+1.4142,\ +1.0000)$
$(|\mathbf{a}|,\ \mathbf{b} \cdot \mathbf{q}) \to (+1.4142,\ +1.0000)$
$(|\mathbf{a}|,\ \mathbf{c} \cdot \mathbf{p}) \to (+1.7321,\ +1.0000)$
$(|\mathbf{a}|,\ \mathbf{c} \cdot \mathbf{q}) \to (+1.7321,\ +1.0000)$
$(|\mathbf{b}|,\ \mathbf{a} \cdot \mathbf{p}) \to (+1.7321,\ +1.0000)$
$(|\mathbf{b}|,\ \mathbf{a} \cdot \mathbf{q}) \to (+1.7321,\ +1.0000)$
$(|\mathbf{b}|,\ \mathbf{c} \cdot \mathbf{p}) \to (+1.7321,\ -1.0000)$
$(|\mathbf{b}|,\ \mathbf{c} \cdot \mathbf{q}) \to (+1.7321,\ -1.0000)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +9.7566, +0.0000, +42.5959, +0.0000, +105.0660, +0.0000, +154.3837, +0.0000, +128.3933, +0.0000, +48.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{a} \cdot \mathbf{p}) \to (+1.7321,\ -1.0000)$
$(|\mathbf{a}|,\ \mathbf{a} \cdot \mathbf{q}) \to (+1.7321,\ -1.0000)$
$(|\mathbf{b}|,\ \mathbf{b} \cdot \mathbf{p}) \to (+1.4142,\ -1.0000)$

$(|\mathbf{b}|,\ \mathbf{b}\cdot\mathbf{q})\rightarrow(+1.7321,\ +1.0000)$
$(|\mathbf{c}|,\ \mathbf{c}\cdot\mathbf{p})\rightarrow(+1.7321,\ +1.0000)$
$(|\mathbf{c}|,\ \mathbf{c}\cdot\mathbf{q})\rightarrow(+1.4142,\ +1.0000)$

4.12 Block: $|\mathbf{x}|\times\mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0,0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: +4.8783, +3.0000, +5.8990, +9.7566, +0.7785, +7.8990
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \mathbf{b}\cdot\mathbf{c})\rightarrow(+1.7321,\ +2.0000)$
$(|\mathbf{b}|,\ \mathbf{a}\cdot\mathbf{c})\rightarrow(+1.4142,\ +1.0000)$
$(|\mathbf{c}|,\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.7321,\ +0.0000)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +9.7566, +6.0000, +26.5959, +48.7832, +1.2071, +146.3837, -96.9898, +198.7540, -140.8499, +114.7878, -60.9898, +17.6705
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.7321,\ +2.0000)$
$(|\mathbf{a}|,\ \mathbf{a}\cdot\mathbf{c})\rightarrow(+1.4142,\ +2.0000)$
$(|\mathbf{b}|,\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.7321,\ +1.0000)$
$(|\mathbf{b}|,\ \mathbf{b}\cdot\mathbf{c})\rightarrow(+1.7321,\ +1.0000)$
$(|\mathbf{c}|,\ \mathbf{a}\cdot\mathbf{c})\rightarrow(+1.7321,\ +0.0000)$
$(|\mathbf{c}|,\ \mathbf{b}\cdot\mathbf{c})\rightarrow(+1.4142,\ +0.0000)$

4.13 Block: $|\mathbf{x}|\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +19.5133, +174.3837, +943.7509, +3444.8245, +8934.3634, +16882.2855, +23417.7226, +23665.7018, +16992.6589, +8228.7183, +2412.9023, +324.0000
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ +0.0000)$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ +0.0000)$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ +0.0000)$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ -0.0000)$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ -0.0000)$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ -0.0000)$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ +0.0000)$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ +0.0000)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(1,0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: +9.7566, +39.5959, +85.5527, +103.7878, +67.0251, +18.0000
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ +0.0000)$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ +0.0000)$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ +0.0000)$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow(+1.7321,\ -0.0000)$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow(+1.4142,\ -0.0000)$
$(|\mathbf{c}|,\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(+1.4142,\ -0.0000)$

4.14 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +19.5133, +186.3837, +1141.4262, +4963.5102, +16082.9329, +39708.0812, +75123.8548, +107957.5499, +114861.2004, +85907.3300, +40614.0886, +9216.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))\rightarrow (+1.4142,\ -2.0000)$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))\rightarrow (+1.4142,\ -2.0000)$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))\rightarrow (+1.4142,\ +2.0000)$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))\rightarrow (+1.4142,\ +2.0000)$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))\rightarrow (+1.7321,\ +1.0000)$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))\rightarrow (+1.7321,\ +1.0000)$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))\rightarrow (+1.7321,\ +1.0000)$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))\rightarrow (+1.7321,\ +1.0000)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +19.5133, +0.0000, +186.3837, +0.0000, +1137.6121, +0.0000, +4907.0857, -0.0000, +15685.5403, -0.0000, +37980.0812, -0.0000, +70047.4790, +0.0000, +97562.9173, +0.0000, +100076.0612, +0.0000, +71849.2564, +0.0000, +32509.9005, +0.0000, +7056.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))\rightarrow (+1.7321,\ -2.0000)$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))\rightarrow (+1.7321,\ -2.0000)$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))\rightarrow (+1.7321,\ +2.0000)$
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))\rightarrow (+1.7321,\ +2.0000)$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))\rightarrow (+1.7321,\ -1.0000)$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))\rightarrow (+1.7321,\ -1.0000)$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))\rightarrow (+1.4142,\ -1.0000)$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))\rightarrow (+1.4142,\ -1.0000)$
... (4 more pairs)

4.15 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: +9.7566, +51.5959, +163.6057, +342.1714, +429.5775, +294.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))\rightarrow (+1.7321,\ -2.0000)$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))\rightarrow (+1.7321,\ -2.0000)$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))\rightarrow (+1.4142,\ -2.0000)$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))\rightarrow (+1.7321,\ +2.0000)$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))\rightarrow (+1.7321,\ +2.0000)$
$(|\mathbf{c}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))\rightarrow (+1.4142,\ +2.0000)$

4.16 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(0, 2)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{p} \cdot \mathbf{q})\rightarrow (-1.0000,\ -1.0000)$

4.17 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: -6.0000, +0.0000, +18.0000, +0.0000, -32.0000, +0.0000, +36.0000, +0.0000, -24.0000, +0.0000, +8.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{p}) \to (-1.0000,\ +1.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{q}) \to (-1.0000,\ +1.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{p}) \to (-1.0000,\ +1.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{q}) \to (-1.0000,\ -1.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{p}) \to (-1.0000,\ -1.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{q}) \to (-1.0000,\ -1.0000)$

4.18 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: -3.0000, +3.0000, +1.0000, -6.0000, +1.0000, +3.0000
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{b}) \to (-1.0000,\ +2.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{c}) \to (-1.0000,\ +1.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{c}) \to (-1.0000,\ +0.0000)$

4.19 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 2)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: -6.0000, +15.0000, -20.0000, +15.0000, -6.0000, +1.0000
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ -0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ -0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ -0.0000)$

4.20 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: -12.0000, +78.0000, -340.0000, +1089.0000, -2664.0000, +5072.0000, -7536.0000, +8664.0000, -7520.0000, +4704.0000, -1920.0000, +400.0000
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-1.0000,\ -2.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (-1.0000,\ -2.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (-1.0000,\ +2.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (-1.0000,\ +2.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (-1.0000,\ +1.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (-1.0000,\ +1.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-1.0000,\ +1.0000)$

$(\mathbf{p}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-1.0000,\ +1.0000)$
... (4 more pairs)

4.21 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= $(0,0)$, type=12, kind=ASSOC, encoded dim= 2
Encoded values: -2.0000, +5.0000
Decoded pairs (u,v):
$(\mathbf{p}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-1.0000,\ -2.0000)$
$(\mathbf{p}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-1.0000,\ +2.0000)$

4.22 Block: $\mathbf{x}\cdot\mathbf{y}\times\mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=11_sigma, kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +12.0000, +0.0000, +60.0000, +0.0000, +160.0000, +0.0000, +240.0000, +0.0000, +192.0000, +0.0000, +64.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{q})\rightarrow(+1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{p})\rightarrow(+1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{c}\cdot\mathbf{q})\rightarrow(+1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{c}\cdot\mathbf{p})\rightarrow(+1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{q})\rightarrow(+1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{p})\rightarrow(+1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{c}\cdot\mathbf{q})\rightarrow(-1.0000,\ +1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{c}\cdot\mathbf{p})\rightarrow(-1.0000,\ +1.0000)$
... (4 more pairs) PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0,1)$, type=11_sigma, kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, -12.0000, +0.0000, +60.0000, +0.0000, -160.0000, +0.0000, +240.0000, +0.0000, -192.0000, +0.0000, +64.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{p})\rightarrow(-1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{q})\rightarrow(-1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{c}\cdot\mathbf{p})\rightarrow(-1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{c}\cdot\mathbf{q})\rightarrow(-1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{p})\rightarrow(-1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{q})\rightarrow(-1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{c}\cdot\mathbf{p})\rightarrow(+1.0000,\ +1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{c}\cdot\mathbf{q})\rightarrow(+1.0000,\ +1.0000)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1,0)$, type=11_sigma, kind=ASSOC, encoded dim= 6
Encoded values: +0.0000, +6.0000, +0.0000, +12.0000, +0.0000, +8.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{q})\rightarrow(+1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{p})\rightarrow(+1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{q})\rightarrow(+1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{p})\rightarrow(-1.0000,\ +1.0000)$
$(\mathbf{c}\cdot\mathbf{p},\ \mathbf{c}\cdot\mathbf{q})\rightarrow(-1.0000,\ +1.0000)$
$(\mathbf{c}\cdot\mathbf{q},\ \mathbf{c}\cdot\mathbf{p})\rightarrow(-1.0000,\ +1.0000)$
PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1,1)$, type=null_self, kind=NULL_SELF, encoded dim= 0
Decoded pairs (u,v):

$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{p})\rightarrow(-1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{q})\rightarrow(-1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{p})\rightarrow(-1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{q})\rightarrow(+1.0000,\ +1.0000)$
$(\mathbf{c}\cdot\mathbf{p},\ \mathbf{c}\cdot\mathbf{p})\rightarrow(+1.0000,\ +1.0000)$
$(\mathbf{c}\cdot\mathbf{q},\ \mathbf{c}\cdot\mathbf{q})\rightarrow(+1.0000,\ +1.0000)$

4.23 Block: $\mathbf{x}\cdot\mathbf{y}\times\mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +6.0000, -16.0000, +0.0000, +0.0000, -24.0000, +25.0000, +0.0000, +0.0000, +18.0000, -10.0000, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{c})\rightarrow(+1.0000,\ +2.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{c})\rightarrow(-1.0000,\ +2.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{c})\rightarrow(+1.0000,\ +1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{c})\rightarrow(+1.0000,\ +0.0000)$
$(\mathbf{c}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{b})\rightarrow(-1.0000,\ +1.0000)$
$(\mathbf{c}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{b})\rightarrow(-1.0000,\ +0.0000)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +12.0000, -68.0000, +0.0000, +0.0000, -240.0000, +594.0000, +0.0000, +0.0000, +1104.0000, -1612.0000, +0.0000, +0.0000, -1896.0000, +1809.0000, +0.0000, +0.0000, +1380.0000, -824.0000, +0.0000, +0.0000, -360.0000, +100.0000, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.0000,\ +2.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.0000,\ +2.0000)$
$(\mathbf{a}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{c})\rightarrow(-1.0000,\ +2.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{c})\rightarrow(-1.0000,\ +2.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{a}\cdot\mathbf{b})\rightarrow(-1.0000,\ +1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{a}\cdot\mathbf{b})\rightarrow(-1.0000,\ +1.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{c})\rightarrow(-1.0000,\ +0.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{c})\rightarrow(-1.0000,\ +0.0000)$
... (4 more pairs)

4.24 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0,1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, -6.0000, +0.0000, +0.0000, +0.0000, +15.0000, +0.0000, +0.0000, +0.0000, -20.0000, +0.0000, +0.0000, +0.0000, +15.0000, +0.0000, +0.0000, +0.0000, -6.0000, +0.0000, +0.0000, +0.0000, +1.0000, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow(-1.0000,\ +0.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow(-1.0000,\ +0.0000)$
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(-1.0000,\ +0.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(-1.0000,\ -0.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow(-1.0000,\ -0.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow(-1.0000,\ -0.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(+1.0000,\ +0.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(+1.0000,\ +0.0000)$
... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap=

$(1,1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.0000, -3.0000, +0.0000, +0.0000, +0.0000, +3.0000, +0.0000, +0.0000, +0.0000, -1.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-1.0000,\ +0.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-1.0000,\ +0.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(-1.0000,\ +0.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(+1.0000,\ -0.0000)$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+1.0000,\ -0.0000)$
$(\mathbf{c}\cdot\mathbf{q},\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+1.0000,\ -0.0000)$

4.25 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +6.0000, +0.0000, +33.0000, +0.0000, +48.0000, +0.0000, +216.0000, +0.0000, +96.0000, +0.0000, +400.0000
Decoded pairs (u, v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(-1.0000,\ -2.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+1.0000,\ -2.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to(-1.0000,\ +2.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to(+1.0000,\ +2.0000)$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(-1.0000,\ -1.0000)$
$(\mathbf{c}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(-1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{p},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+1.0000,\ -1.0000)$
... (4 more pairs) PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +6.0000, +0.0000, +33.0000, +0.0000, +48.0000, +0.0000, +216.0000, +0.0000, +96.0000, +0.0000, +400.0000
Decoded pairs (u, v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(-1.0000,\ -2.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+1.0000,\ -2.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to(-1.0000,\ +2.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to(+1.0000,\ +2.0000)$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(-1.0000,\ -1.0000)$
$(\mathbf{c}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(-1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{p},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+1.0000,\ -1.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+1.0000,\ -1.0000)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +6.0000, +0.0000, +0.0000, +0.0000, +33.0000, +0.0000, +0.0000, +0.0000, +48.0000, +0.0000, +0.0000, +0.0000, +216.0000, +0.0000, +0.0000, +0.0000, +96.0000, +0.0000, +0.0000, +0.0000, +400.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(-1.0000,\ -2.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(+1.0000,\ -2.0000)$
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to(-1.0000,\ +2.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to(+1.0000,\ +2.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(-1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(-1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+1.0000,\ -1.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+1.0000,\ -1.0000)$

... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24

Encoded values: +0.0000, +0.0000, +6.0000, +0.0000, +0.0000, +0.0000, +33.0000, +0.0000, +0.0000, +0.0000, +48.0000, +0.0000, +0.0000, +0.0000, +216.0000, +0.0000, +0.0000, +0.0000, +96.0000, +0.0000, +0.0000, +0.0000, +400.0000, +0.0000

Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{p},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-1.0000,\ -2.0000)$
$(\mathbf{a} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (+1.0000,\ -2.0000)$
$(\mathbf{a} \cdot \mathbf{p},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (-1.0000,\ +2.0000)$
$(\mathbf{a} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (+1.0000,\ +2.0000)$
$(\mathbf{b} \cdot \mathbf{p},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-1.0000,\ -1.0000)$
$(\mathbf{b} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (-1.0000,\ -1.0000)$
$(\mathbf{b} \cdot \mathbf{p},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (+1.0000,\ -1.0000)$
$(\mathbf{b} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (+1.0000,\ -1.0000)$
... (4 more pairs)

4.26 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12

Encoded values: +0.0000, +18.0000, +0.0000, +183.0000, +0.0000, +1116.0000, +0.0000, +4575.0000, +0.0000, +11250.0000, +0.0000, +15625.0000

Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{p},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ -2.0000)$
$(\mathbf{a} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ -2.0000)$
$(\mathbf{b} \cdot \mathbf{p},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ -2.0000)$
$(\mathbf{b} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (-1.0000,\ -2.0000)$
$(\mathbf{c} \cdot \mathbf{p},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (-1.0000,\ -2.0000)$
$(\mathbf{c} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (-1.0000,\ -2.0000)$
$(\mathbf{a} \cdot \mathbf{p},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ +2.0000)$
$(\mathbf{a} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ +2.0000)$
... (4 more pairs)

4.27 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6

Encoded values: +8.4853, +32.0000, +67.8823, +85.0000, +59.3970, +20.0000

Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \mathbf{a} \cdot \mathbf{c}) \to (+0.0000,\ +2.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{b}) \to (+1.0000,\ +2.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ \mathbf{b} \cdot \mathbf{c}) \to (+2.0000,\ +1.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{b}) \to (+2.0000,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \mathbf{b} \cdot \mathbf{c}) \to (+0.0000,\ +1.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{c}) \to (+1.0000,\ +0.0000)$
PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0

Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \mathbf{a} \cdot \mathbf{b}) \to (+0.0000,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{c}) \to (+1.0000,\ +1.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \mathbf{b} \cdot \mathbf{c}) \to (+2.0000,\ +2.0000)$

4.28 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: +6.0000, +13.0000, +12.0000, +4.0000, +0.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+2.0000,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+2.0000,\ +0.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+1.0000,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+1.0000,\ -0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$
PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +12.0000, +62.0000, +180.0000, +321.0000, +360.0000, +248.0000, +96.0000, +16.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+2.0000,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+2.0000,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+2.0000,\ +0.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+2.0000,\ -0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+1.0000,\ -0.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+1.0000,\ -0.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+1.0000,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+1.0000,\ +0.0000)$
... (4 more pairs)

4.29 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +12.0000, +0.0000, +74.0000, +0.0000, +300.0000, +0.0000, +897.0000, +0.0000, +2088.0000, +0.0000, +3868.0000, +0.0000, +5688.0000, +0.0000, +6612.0000, +0.0000, +6000.0000, +0.0000, +4208.0000, +0.0000, +2112.0000, +0.0000, +640.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (+2.0000,\ -2.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (+0.0000,\ -2.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (+2.0000,\ +2.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (+0.0000,\ +2.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (+2.0000,\ +1.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (+2.0000,\ -1.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (+0.0000,\ -1.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (+1.0000,\ -1.0000)$
... (4 more pairs) PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(2, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +12.0000, +74.0000, +300.0000, +879.0000, +1944.0000, +3340.0000, +4536.0000, +4959.0000, +4380.0000, +3050.0000, +1500.0000, +625.0000
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (+1.0000,\ -2.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (+1.0000,\ -2.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (+2.0000,\ -1.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (+2.0000,\ -1.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (+2.0000,\ +1.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (+2.0000,\ +1.0000)$

$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(+1.0000,\ +2.0000)$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(+1.0000,\ +2.0000)$
... (4 more pairs)

4.30 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(2,0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: +6.0000, +25.0000, +60.0000, +124.0000, +144.0000, +160.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+2.0000,\ +2.0000)$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+0.0000,\ +2.0000)$
$(\mathbf{b}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+1.0000,\ +2.0000)$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+2.0000,\ -2.0000)$
$(\mathbf{a}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+0.0000,\ -2.0000)$
$(\mathbf{b}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+1.0000,\ -2.0000)$

4.31 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=neg_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(0,2)$, type=`neg_sigma`, kind=`ASSOC`, encoded dim= 6
Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
... (4 more pairs) PairFlavour overlap=(1, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1,2)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$
$(-\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$

4.32 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +12.0000, +0.0000, +54.0000, +0.0000, +116.0000, +0.0000, +129.0000, +0.0000, +72.0000, +0.0000, +16.0000
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+0.0000,\ +2.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+0.0000,\ +2.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to(+0.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to(-0.0000,\ -2.0000)$
$(\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(-0.0000,\ -1.0000)$

$(-\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-0.0000,\ -1.0000)$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ -1.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+0.0000,\ -1.0000)$
... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +12.0000, +0.0000, +0.0000, +0.0000, +54.0000, +0.0000, +0.0000, +0.0000, +116.0000, +0.0000, +0.0000, +0.0000, +129.0000, +0.0000, +0.0000, +0.0000, +72.0000, +0.0000, +0.0000, +0.0000, +16.0000, +0.0000
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.0000,\ +2.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.0000,\ +2.0000)$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(-0.0000,\ -2.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-0.0000,\ -1.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-0.0000,\ -1.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ -1.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+0.0000,\ -1.0000)$
... (4 more pairs)

4.33 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=22 kind=ASSOC dim=6 **PairFlavour** overlap= $(1,0)$, type=`22`, kind=`ASSOC`, encoded dim= 6
Encoded values: -24.0000, +240.0000, -1280.0000, +3840.0000, -6144.0000, +4096.0000
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ +2.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ +2.0000)$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.0000,\ -2.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.0000,\ +2.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.0000,\ +2.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -2.0000)$
... (4 more pairs)

4.34 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +0.0000, +4.0000, +0.0000, +0.0000, +10.0000, +0.0000, +0.0000, +0.0000, +0.0000, +56.0000, +0.0000, +0.0000, +569.0000, +0.0000, +0.0000, +0.0000, +0.0000, +2500.0000, +0.0000, +0.0000, -2500.0000, +0.0000
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(-2.0000,\ -1.0000)$
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(-1.0000,\ -2.0000)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-2.0000,\ +1.0000)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-1.0000,\ +2.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+1.0000,\ +2.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+2.0000,\ +1.0000)$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-1.0000,\ +1.0000)$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-1.0000,\ +1.0000)$
... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +0.0000, -4.0000, +0.0000, +0.0000, +10.0000, +0.0000, +0.0000,

+0.0000, +0.0000, -56.0000, +0.0000, +0.0000, +569.0000, +0.0000, +0.0000, +0.0000, +0.0000, -2500.0000, +0.0000, +0.0000, -2500.0000, +0.0000
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (-2.0000,\ +1.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (-1.0000,\ +2.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-2.0000,\ -1.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (-1.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (+1.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (+2.0000,\ -1.0000)$
$(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-1.0000,\ -1.0000)$
$(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (-1.0000,\ -1.0000)$
... (4 more pairs) PairFlavour overlap=(2, 0) type=neg_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(2, 0)$, type=`neg_sigma`, kind=`ASSOC`, encoded dim= 6
Encoded values: -24.0000, -216.0000, +928.0000, +2064.0000, -2304.0000, -1024.0000
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (+2.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (-2.0000,\ +2.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (+2.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-2.0000,\ +2.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (+1.0000,\ -1.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}),\ -\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (-1.0000,\ +1.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (+1.0000,\ -1.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (-1.0000,\ +1.0000)$
... (4 more pairs) PairFlavour overlap=(2, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (+1.0000,\ +1.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (+1.0000,\ +1.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (+1.0000,\ +1.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (+1.0000,\ +1.0000)$
$(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}),\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (+2.0000,\ +2.0000)$
$(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}),\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (+2.0000,\ +2.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-1.0000,\ -1.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (-1.0000,\ -1.0000)$
... (4 more pairs)

4.35 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=neg kind=ASSOC dim=12 **PairFlavour** overlap= $(2, 0)$, type=`neg`, kind=`ASSOC`, encoded dim= 12
Encoded values: -12.0000, +0.0000, +150.0000, +0.0000, -1068.0000, +0.0000, +6129.0000, +0.0000, -19200.0000, +0.0000, +40000.0000, +0.0000
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+2.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (-2.0000,\ +2.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+2.0000,\ +2.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (-2.0000,\ -2.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (-1.0000,\ +2.0000)$
$(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}),\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ -2.0000)$
$(-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}),\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (-1.0000,\ +2.0000)$
... (4 more pairs)

4.36 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z}) \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(3, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(3, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c})) \to (+2.0000,\ +2.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c})) \to (-2.0000,\ -2.0000)$

5 Alignment Decoder (recovers the G-orbit of repS evaluations)

—repS—: $|repS| = 8$

—G—: $|G| = 12$

Decoded vectors: decoded column vectors: 12

Decoded alignment table (rows=repS atoms, cols=group elements): **Decoded alignment table** (rows= repS atoms, columns= $g \in G$):

Table 1: Decoded alignment table

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	
$\lvert\mathbf{a}\rvert$	+1.7321	+1.7321	+1.4142	+1.4142	+1.4142	+1.4142	+1.4142	+1.4142	+1.4142	+1.4
$\lvert\mathbf{p}\rvert$	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0
$\mathbf{a}\cdot\mathbf{b}$	+1.0000	+1.0000	+0.0000	+0.0000	+2.0000	+2.0000	+0.0000	+0.0000	+2.0000	+2.0
$\mathbf{a}\cdot\mathbf{p}$	-1.0000	+1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0
$\mathbf{p}\cdot\mathbf{q}$	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0
$\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c})$	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0000	-2.0
$\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})$	-2.0000	-2.0000	-1.0000	-1.0000	-1.0000	-1.0000	+1.0000	+1.0000	+1.0000	+1.0
$\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})$	+0.0000	+0.0000	-0.0000	-0.0000	-0.0000	-0.0000	+0.0000	+0.0000	+0.0000	+0.0

Ground truth (direct evaluation of g·repS on event): **Ground truth** (direct evaluation of $g \cdot$ repS on event):

Table 2: Ground truth alignment table

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	
$\lvert\mathbf{a}\rvert$	+1.7321	+1.7321	+1.7321	+1.7321	+1.4142	+1.4142	+1.4142	+1.4142	+1.7321	+1.7
$\lvert\mathbf{p}\rvert$	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0
$\mathbf{a}\cdot\mathbf{b}$	+2.0000	+2.0000	+1.0000	+1.0000	+2.0000	+2.0000	+0.0000	+0.0000	+1.0000	+1.0
$\mathbf{a}\cdot\mathbf{p}$	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0
$\mathbf{p}\cdot\mathbf{q}$	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0
$\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c})$	+2.0000	+2.0000	-2.0000	-2.0000	-2.0000	-2.0000	+2.0000	+2.0000	+2.0000	+2.0
$\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})$	-1.0000	+1.0000	+2.0000	-2.0000	+1.0000	-1.0000	+1.0000	-1.0000	-2.0000	+2.0
$\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0

Multiset comparison (decoded vs ground truth): **Multiset comparison:**
MISMATCH — decoded multiset differs from ground truth
Max —error— per repS row (decoded vs ground truth): **Max |error| per repS row:**
Mymag(a): $|\mathbf{a}|$: max err $= 3.18e-01$

Mymag(p): $|\mathbf{p}|$: max err $= 0.00e + 00$

dot(a,b): $\mathbf{a} \cdot \mathbf{b}$: max err $= 0.00e + 00$

dot(a,p): $\mathbf{a} \cdot \mathbf{p}$: max err $= 2.00e + 00$

dot(p,q): $\mathbf{p} \cdot \mathbf{q}$: max err $= 0.00e + 00$

eps(a,b,c): $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$: max err $= 4.00e + 00$

eps(a,b,p): $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$: max err $= 0.00e + 00$

eps(a,p,q): $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$: max err $= 0.00e + 00$

6 Summary

Phase 1 encoded length: Phase 1 encoded length: 25 reals

Phase 2 encoded length: Phase 2 encoded length: 624 reals

Total encoded length: Total encoded length: 649 reals

repS size: |repS| = 8 atoms

—G—: $|G| = 12$ group elements

Decoded vectors: decoded alignment vectors: 12

TeX output written to: /Users/lester/github/Echinocoder/symcoder/demo_shadow_event_parity_-1.tex Compile with: pdflatex demo_shadow_event_parity_-1.tex